

Windows Security Log Investigation & Analysis

Student Name: Ayesha Sajid

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1. Project Overview

This project demonstrates the ability to configure Windows Security Auditing, simulate unauthorized access, and analyze system logs using the Windows Event Viewer. In a Security Operations Center (SOC) environment, these logs are the primary source of truth for detecting brute-force attacks and lateral movement.

2. Technical Objectives

- Configure **Local Security Policies** to enable auditing.
- Generate and detect **Event ID 4625** (Failed Logons).
- Generate and detect **Event ID 4688** (Process Creation).
- Analyze event metadata to identify the source and target of suspicious activity.

3. Implementation Steps & Evidence

Phase 1: Enabling the "Security Camera"

Before investigation, the system must be told which activities to record. I navigated to the Local Security Policy to enable logging for all logon attempts.

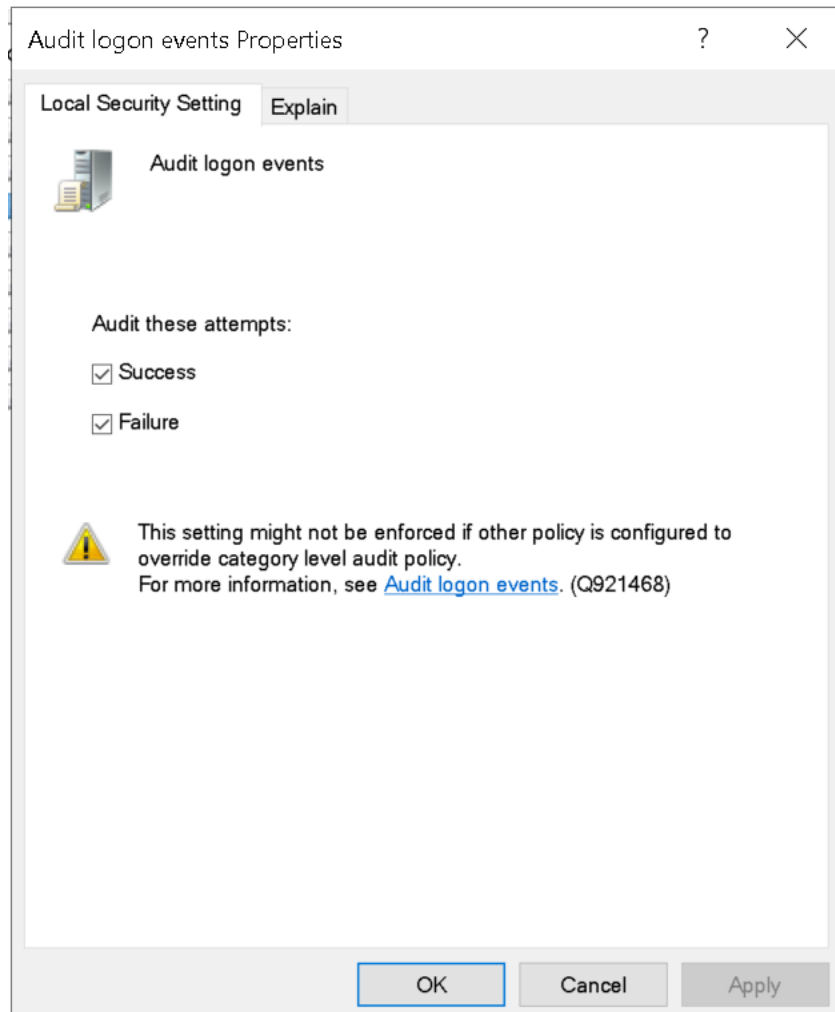


Figure 1: *Configuring Audit logon events for both Success and Failure.*

Phase 2: Detecting Brute-Force Attempts

I simulated a brute-force attack by attempting to log in with a non-existent account. I then used Event Viewer to filter for **Event ID 4625**.

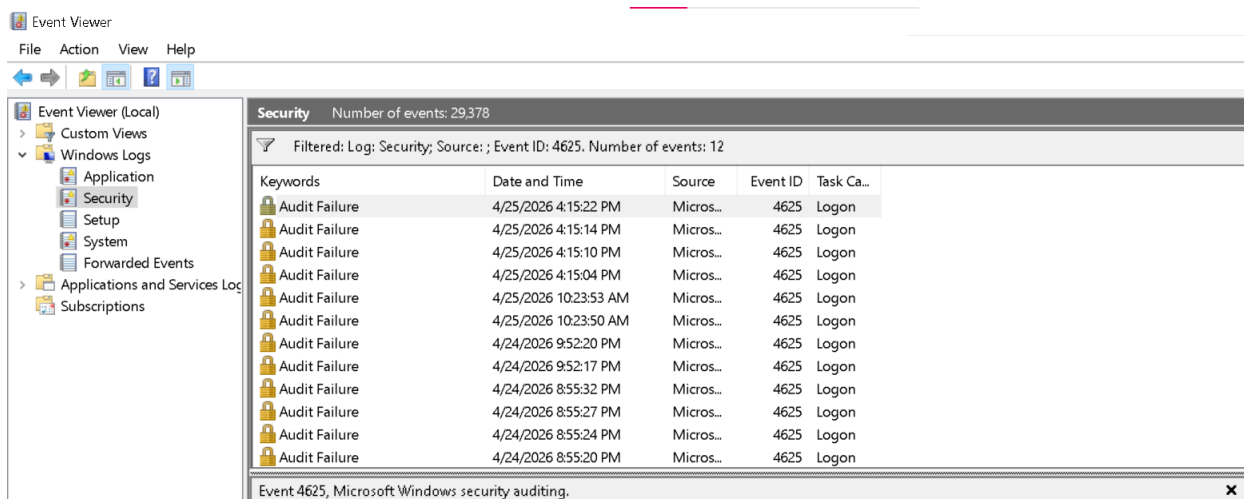


Figure 2: Filtered view of the Security Log showing multiple Audit Failures (Event ID 4625).

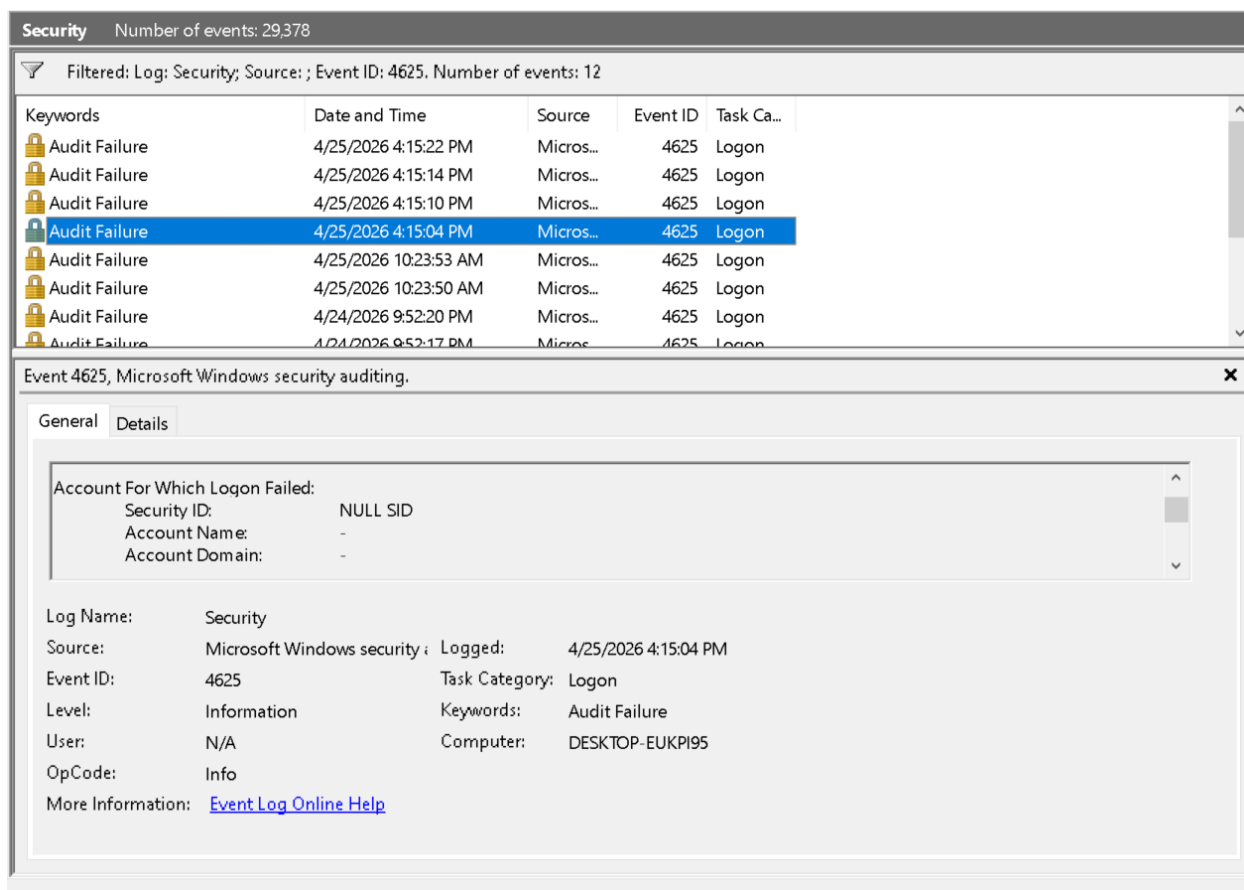


Figure 3: General view of a specific logon failure event before deep-dive analysis.

Phase 3: Incident Analysis (The "Smoking Gun")

By scrolling through the event details, I was able to identify the specific account targeted during the attack. This is a critical skill for identifying if an attacker is targeting high-value accounts (like Admin).

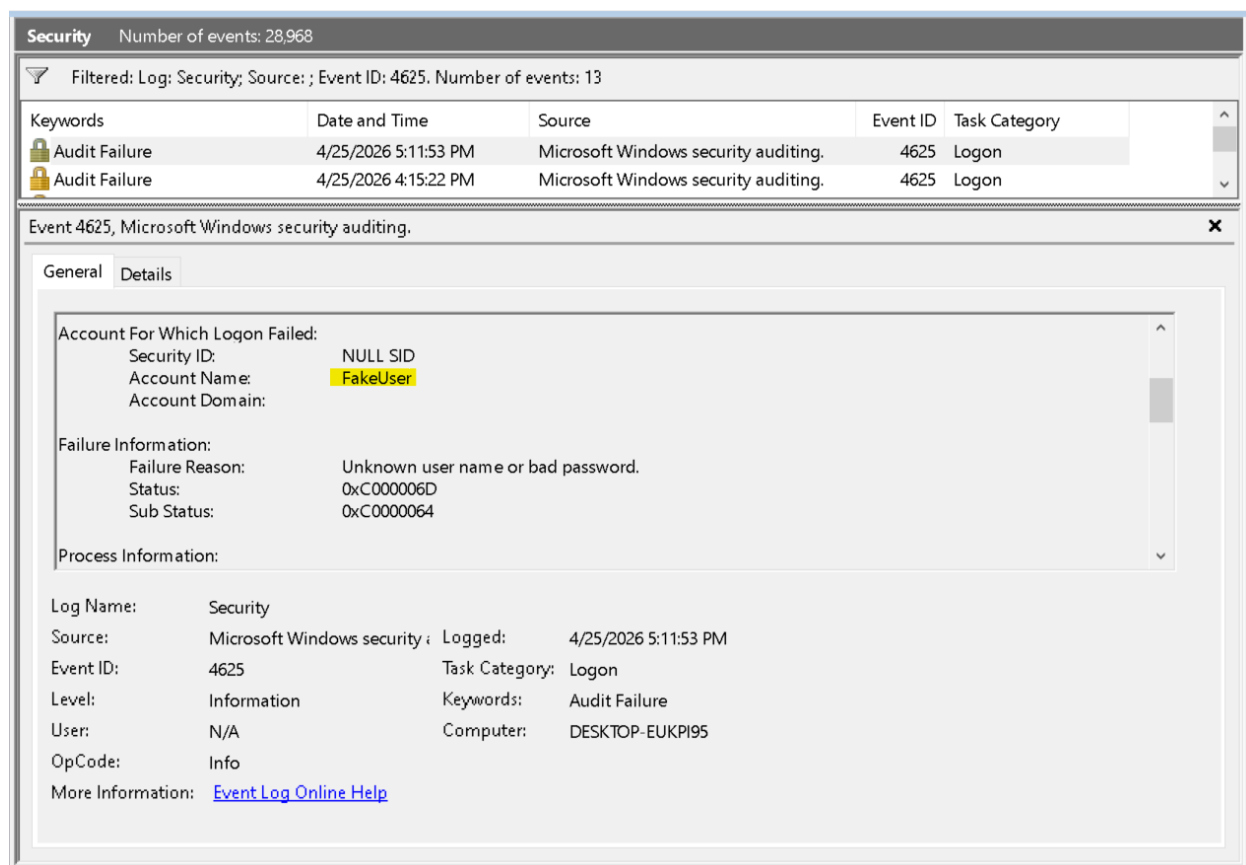


Figure 4: Detailed analysis identifying "FakeUser" as the targeted account and Status Code 0xC000006D.

Phase 4: Monitoring Command Execution

Once an attacker gains access, they typically run reconnaissance commands. I enabled advanced process auditing to track what programs are being executed on the system.

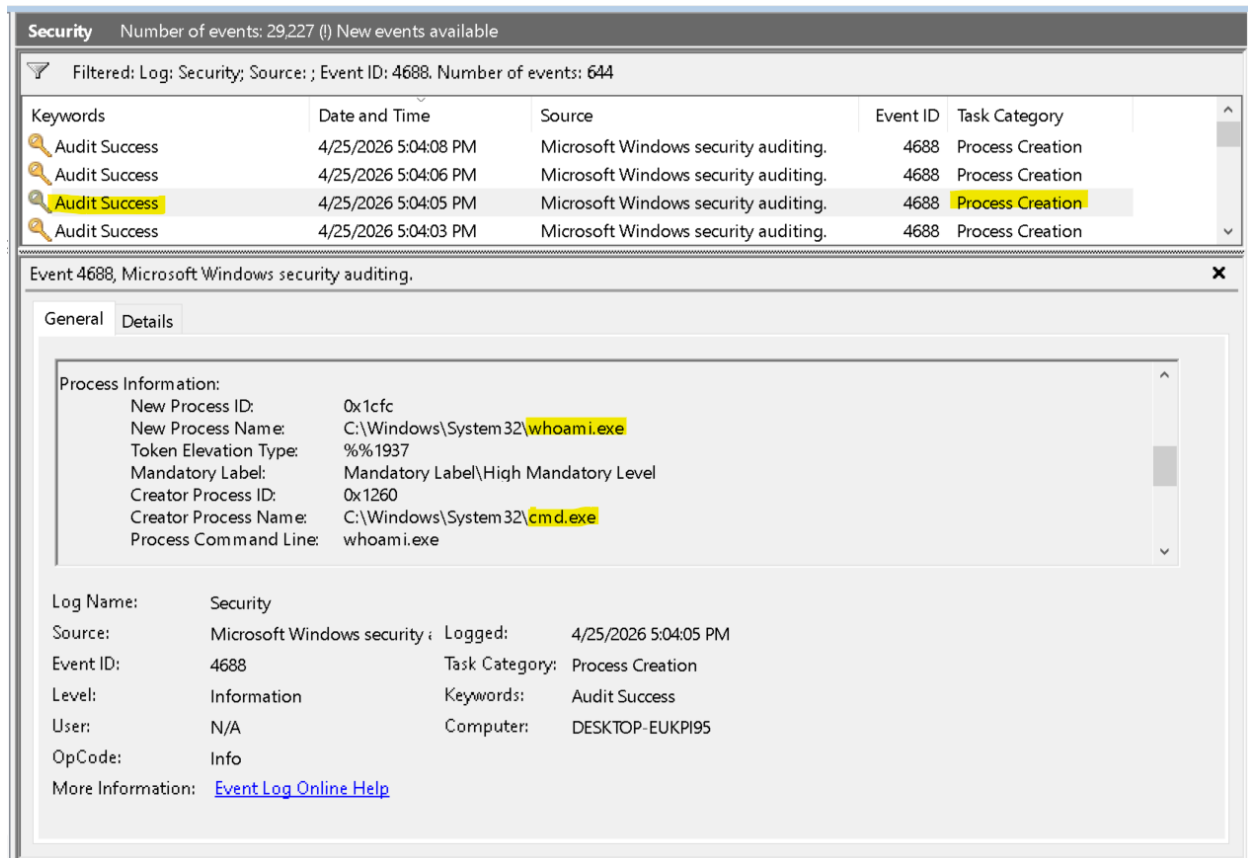


Figure 5: Identification of Event ID 4688. The log clearly shows the execution of *whoami.exe* and the parent process *cmd.exe*.

4. Analysis & Conclusion

During this lab, I successfully identified two major stages of a cyber-attack:

1. **Credential Access:** The system logged unauthorized attempts to access the "FakeUser" account.
 2. **Discovery:** The system logged the use of command-line tools used for reconnaissance.
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